



Carbon Farming for Sustainable Agriculture: A Critical Review of Soil Carbon Sequestration, Nutrient Management and Climate Resilience

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Abstract

Carbon farming has moved from a soil-management concept to a prominent climate-policy proposition, yet the agronomic value of building soil organic carbon and the credibility of soil carbon as a durable climate-removal pathway are not identical questions. This critical narrative review evaluates carbon farming as an integrated strategy for soil carbon sequestration, nutrient management and climate resilience. The literature was selected from accessible scholarly indexes and disciplinary databases, supplemented by citation chaining and authoritative source verification, with emphasis on field experiments, long-term trials, meta-analyses, mechanistic soil-carbon studies and work on measurement, reporting and verification. Evidence indicates that practices which increase carbon inputs, reduce avoidable carbon losses and improve biological nutrient cycling can rebuild

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depleted soil organic carbon, particularly where initial stocks are low and management constraints are substantial. Cover crops, diversified rotations, organic amendments, agroforestry and context-appropriate biochar generally show stronger or more interpretable carbon benefits than simple claims based on reduced tillage alone. Benefits for nutrient retention, aggregation, rooting conditions and yield stability are plausible and often observed, but vary with climate, soil texture and mineralogy, baseline fertility, water regime, residue availability and the duration of management. Carbon gains are finite, reversible and vulnerable to sampling artefacts; transferring manure or crop residues between locations can also create apparent project gains without equivalent net atmospheric removal. Credible carbon farming therefore requires whole-system greenhouse-gas accounting, explicit treatment of additionality and permanence, consistent soil-depth and bulk-density protocols, and conservative crediting. The strongest case for carbon farming is consequently not as a universal substitute for emissions reduction, but as a geographically targeted form of soil restoration that can deliver climate mitigation alongside agronomic and resilience co-benefits when carbon accounting is rigorous and nutrient trade-offs are managed.

Keywords: Soil organic carbon; regenerative agriculture; cover crops; organic amendments; biochar; nutrient cycling; drought resilience; carbon markets.

1. Introduction

Agricultural soils occupy an unusual position in climate and sustainability policy. They are simultaneously a production medium, a major terrestrial carbon reservoir, a source of carbon dioxide and nitrous oxide when poorly managed, and a potential sink when management increases persistent organic carbon stocks. Historical conversion of natural ecosystems to cropland and grazing land has created a substantial soil carbon debt, although the magnitude and spatial pattern of that debt remain uncertain because reconstructions depend on land-use histories, sparse observations and model assumptions (Sanderman *et al.*, 2017). This depletion creates a restoration opportunity: rebuilding soil organic carbon (SOC) can remove atmospheric carbon dioxide while improving properties that matter to agriculture, including aggregation, cation exchange, nutrient buffering and biological activity (Lal, 2004; Paustian *et al.*, 2016). The attraction of this dual-purpose strategy explains the rapid expansion of the term carbon farming across agronomy, climate policy and voluntary carbon markets.

Carbon farming is best understood as a management approach that seeks to increase net carbon retention in agricultural landscapes while avoiding offsetting greenhouse-gas emissions and maintaining productive land use. It therefore includes practices that increase photosynthetic carbon inputs to soil, transfer organic resources to fields, reduce erosion or decomposition losses, introduce long-lived carbon forms, or protect existing high-carbon soils. The concept is broader than SOC sequestration alone because a climate claim depends on net greenhouse-gas consequences, additionality, duration and leakage. Conversely, an agronomic claim may remain valid even when the climate-removal effect is small. Confusing these two evidentiary questions has produced both exaggerated optimism and overly broad scepticism in the literature (Powlson *et al.*, 2011; Smith, 2016). Recent systematic evidence from European croplands similarly shows that soil organic carbon responses differ among carbon-farming practices and depend on how stock changes and sampling depths are evaluated (Petersson *et al.*, 2025).

The global technical potential is large enough to matter but not large enough to remove the need for rapid fossil-fuel emissions reduction. Spatial analyses suggest that croplands with depleted SOC can provide meaningful sequestration opportunities, especially where yield gaps and historical carbon losses coincide (Amelung *et al.*, 2020; Zomer *et al.*, 2017). Yet global potential estimates depend strongly on the assumed adoption area, achievable rate, depth of accounting, persistence and availability of biomass or nutrients. The influential '4 per 1000' framing helped place soil carbon on the international agenda, but its uniform annual target should not be read as a biophysical law. SOC accumulation slows as soils approach new steady states, and some long-term experiments show that rates sufficient to meet such targets require inputs or management changes that cannot be replicated across large areas (Minasny *et al.*, 2017; Poulton *et al.*, 2018).

Mechanistic understanding has also shifted. Soil organic matter is no longer adequately represented as a simple pool of chemically recalcitrant humus that can be increased indefinitely by adding residues. Persistence emerges from interactions among organic compounds, microbial transformation, aggregates, mineral surfaces, accessibility and environmental constraints (Lehmann & Kleber, 2015; Lehmann *et al.*, 2020). A useful operational distinction separates particulate organic matter, which is relatively responsive to recent plant inputs, from mineral-associated organic matter, which often turns over more slowly but is constrained by mineral surface capacity and nutrient stoichiometry (Lavallee *et al.*, 2020). This framework makes clear why the same practice can generate different SOC responses across soils and why measured total carbon change may conceal shifts in stability. A recent global meta-analysis further showed that fertilisation and tillage can affect soil organic carbon fractions differently, reinforcing the importance of distinguishing changes among carbon pools when evaluating management responses (Liu *et al.*, 2024).

The agronomic argument for carbon farming is equally conditional. Higher SOC often accompanies better soil structure and fertility, but SOC is both a cause and a consequence of productive cropping. Fertiliser inputs can raise SOC by increasing crop growth and residue return, while nutrient deficiencies can limit the microbial conversion of plant carbon into persistent soil organic matter. Organic amendments add both carbon and nutrients, creating stronger SOC responses than mineral fertilisers alone in many settings, yet the atmospheric benefit depends on where the organic material would otherwise have decomposed and on emissions during storage, transport and application (Gross & Glaser, 2021; Liu *et al.*, 2023). Carbon and nutrient management therefore cannot be treated as independent objectives. Recent meta-analytic evidence also indicates that conservation-tillage effects on soil organic carbon sequestration and soil greenhouse-gas fluxes vary across management and environmental conditions, supporting whole-system assessment rather than practice-based assumptions (Meng *et al.*, 2024).

Climate resilience provides a further rationale but demands careful interpretation. SOC can improve aggregate stability, infiltration and some components of water retention, and high-quality soils have been associated with greater yield stability under climate variability (Qiao *et al.*, 2022). Nevertheless, the often-repeated claim that small SOC increases automatically transform plant-available water storage is not supported uniformly across soil textures; the direct contribution can be modest compared with mineral composition and structure (Minasny & McBratney, 2018). Resilience is consequently an emergent system property produced by rooting depth, soil structure, crop diversity, water management and nutrient sufficiency rather than by SOC concentration alone. Meta-analytic evidence from the Indian subcontinent likewise indicates that conservation-agriculture effects on crop yield and soil carbon are influenced by rainfall, soil depth and nutrient conditions, supporting context-specific assessment (Padbhushan *et al.*, 2024).

Recent reviews have addressed individual practices, soil carbon potential, biochar, long-term SOC change and carbon-market integrity. What remains insufficiently integrated is the relationship among three questions that are commonly discussed separately: which management pathways build persistent SOC, how carbon accumulation interacts with nutrient management, and when those changes materially improve resilience while constituting credible climate mitigation. A critical synthesis is needed because evidence from topsoil concentration measurements, field-scale agronomy and carbon-credit protocols is not directly interchangeable. This review therefore evaluates carbon farming as a coupled biophysical, agronomic and governance problem rather than as a list of purportedly 'carbon-positive' practices.

1.1 Scope, Research Gap and Objectives

This review focuses on agricultural mineral soils used for annual and perennial cropping, with selective consideration of agroforestry where it directly informs soil-carbon processes. It evaluates management strategies that alter SOC stocks, nutrient cycling or climate resilience, and it distinguishes soil-quality benefits from verified atmospheric carbon removal. Wetland and peatland conservation, forest carbon projects, livestock-enteric mitigation and industrial carbon capture are outside the central scope except where they affect interpretation of agricultural carbon accounting. The principal objectives are to integrate current understanding of SOC formation and persistence; critically compare the evidence for major carbon-farming practices; examine carbon–nutrient interactions and resilience outcomes; assess methodological and greenhouse-gas accounting limitations; and identify research and governance priorities needed for defensible implementation. The intended contribution is an evidence-calibrated framework that treats carbon farming primarily as soil-system management and only secondarily as a tradable carbon-removal claim when measurement, additionality and permanence requirements are met.

2. Methods for Literature Selection

2.1 Review Design and Rationale

A critical narrative review design was used because the topic spans biophysical mechanisms, agronomic experiments, meta-analyses, modelling and carbon-governance literature with heterogeneous interventions and outcomes that cannot be reduced meaningfully to a single pooled effect. Narrative synthesis permits comparison of evidence at these different levels while retaining attention to mechanism, external validity and conflicting interpretations. Methodological transparency remained essential; the review was structured using principles of explicit justification, literature-search description, scientific reasoning and appropriate referencing consistent with the SANRA framework for narrative reviews (Baethge *et al.*, 2019).

2.2 Information Sources

Literature was identified using Google Scholar, Semantic Scholar, AGRIS, PubMed, PubMed Central, OpenAlex, DOAJ and CORE. Searches were supplemented by backward and forward citation searching from influential reviews and primary studies, and by verification against DOI records, scholarly landing pages and authoritative institutional sources when necessary. The source set was selected to cover soil science, agronomy, biogeochemistry, climate mitigation and environmental policy while retaining access to bibliographic metadata and, where available, abstracts or full text.

2.3 Search Strategy and Search Period

The principal search period was 1 January 2000 to 30 June 2026. Representative concepts were combined using Boolean operators, including ('carbon farming' OR 'soil carbon sequestration' OR 'soil organic carbon') AND (agriculture OR cropland); ('cover crop*' OR rotation OR 'reduced tillage' OR 'no till' OR manure OR compost OR biochar OR agroforestry) AND ('soil organic carbon' OR sequestration); ('soil carbon' AND nitrogen AND phosphorus AND fertilization); ('soil organic carbon' AND drought AND resilience AND yield); and ('soil carbon credit*' OR 'carbon farming') AND (measurement OR verification OR permanence OR additionality OR leakage). Foundational papers within or close to the starting period were retained where they established concepts still central to interpretation. Searches emphasised evidence available up to the final search date of 30 June 2026.

2.4 Eligibility Criteria

Priority was given to peer-reviewed field experiments, long-term trials, meta-analyses, systematic evidence syntheses, mechanistic soil-carbon studies and policy analyses with clear relevance to agricultural SOC, nutrient management, resilience or carbon accounting. Global and multi-site evidence was preferred for claims of generality, whereas individual experiments were used when they illuminated mechanisms or long-term behaviour. English-language publications were included when bibliographic identity and

claim relevance could be verified. Studies were excluded when they relied primarily on laboratory conditions without clear field relevance, treated carbon concentration as equivalent to stock without adequate context, lacked verifiable metadata, focused on unrelated land uses, or provided claims that could not be linked to a stable scholarly source. Preprints and commercial promotional material were not used as evidentiary anchors.

2.5 Screening, Duplicate Handling and Study Selection

Titles and abstracts were assessed for direct relevance to the review questions, followed by examination of full text or sufficiently detailed scholarly records for the sources used to support quantitative or contested claims. Duplicate records appearing across indexes were consolidated by DOI and bibliographic identity. When multiple versions of a paper were found, the version of record was used. Sources with unresolved discrepancies in title, authorship, year or DOI were not retained. No PRISMA-style record counts are reported because the purpose was critical narrative selection rather than exhaustive systematic enumeration.

2.6 Critical Appraisal and Evidence Synthesis

Evidence was appraised according to study design, duration, sample size, spatial coverage, soil-depth treatment, baseline SOC, handling of bulk density and rock fragments, control comparability, greenhouse-gas boundaries, treatment of confounding and consistency with independent studies. Meta-analyses and long-term experiments were given greater weight for broad management claims, while mechanistic papers were used to explain heterogeneity rather than to infer field-scale sequestration rates directly. The synthesis was organised around carbon persistence, management pathways, nutrient coupling, resilience, whole-system climate effects and measurement/reporting/verification. Where studies disagreed, potential reasons such as depth redistribution, carbon-source transfer, climate, soil mineralogy, experimental duration and management intensity were examined before assigning confidence to conclusions.

3. Biophysical Foundations of Carbon Farming

3.1 From Carbon Input to Persistent Soil Organic Carbon

Carbon farming begins with a simple mass-balance proposition: SOC increases only when carbon inputs retained in soil exceed carbon losses over a defined depth and time interval. The processes underlying that balance are not simple. Plant litter, roots, rhizodeposits, manures and amendments enter a soil system in which decomposers transform organic compounds, respire much of the carbon and convert a smaller fraction into microbial products that may become physically or chemically protected. Contemporary soil science therefore places less emphasis on intrinsic molecular recalcitrance and more on accessibility, microbial processing, aggregation and mineral association (Lehmann & Kleber, 2015; Lehmann *et al.*, 2020). This shift matters for management because a practice that increases gross residue input does not necessarily generate proportional long-term storage.

Particulate organic matter (POM) and mineral-associated organic matter (MAOM) provide a useful lens for distinguishing rapid from potentially persistent responses. POM is closely linked to relatively recent plant inputs and can respond quickly to cover crops or residue retention, but it is also vulnerable to disturbance and decomposition. MAOM is formed largely through microbial processing and adsorption or complexation with mineral surfaces; it can be more persistent, yet its formation may be limited by clay mineralogy, reactive surface area and nutrient availability (Lavalley *et al.*, 2020). Carbon-farming interventions should therefore be evaluated not only by total SOC change but, where possible, by the fractions in which new carbon accumulates.

The concept of carbon saturation further limits universal sequestration rates. Soils differ in their capacity to stabilise additional carbon through physical protection and organo-mineral associations. Six *et al.* (2002) formalised this idea by linking stabilisation mechanisms to finite protective capacity. Subsequent work has reinforced that the response to added carbon depends on the distance from a site-specific equilibrium. Depleted croplands may respond strongly at first, whereas soils near a new steady state show declining marginal gains. This non-linearity is a central reason why extrapolating early trial rates over many decades exaggerates cumulative removal.

Depth also changes interpretation. Management can alter the vertical distribution of carbon without substantially changing the stock across the full rooting profile. Reduced tillage is the classic example: surface SOC commonly increases, yet gains may diminish or disappear when deeper layers are included (Powlson *et al.*, 2014). Rooting depth, subsoil carbon inputs and erosion deposition further complicate whole-profile accounting. A climate claim should therefore specify the depth and revisit deeper layers when the management practice changes residue placement, rooting or soil mixing.

SOC persistence is consequently a property of the soil–plant–microbe system rather than a fixed characteristic of a management label. The same cover crop may add carbon but fail to produce durable storage where water limits biomass, while the same manure input may build SOC rapidly in clay-rich soil and less efficiently in coarse soil. The evidence supports process-based targeting: practices should be matched to the limiting mechanism, whether insufficient carbon input, nutrient limitation, erosion, poor aggregation, acidic conditions or lack of mineral protection.

3.2 Carbon Stocks, Sequestration Potential and the Problem of Scale

Global estimates are useful for identifying the order of magnitude of opportunity but can become misleading when interpreted as guaranteed removals. Zomer *et al.* (2017) estimated a substantial technical cropland sequestration potential, with the greatest

opportunities concentrated in intensively cultivated and depleted regions. Bossio *et al.* (2020) likewise identified soil carbon as an important component of natural climate solutions, including both protection of existing stocks and rebuilding of depleted stocks. These analyses support the strategic importance of agricultural soils, but their estimates are scenario-dependent and do not imply that the full biophysical potential can be mobilised economically or credited with high certainty.

The '4 per 1000' initiative illustrates the difference between a motivating global benchmark and a field-scale expectation. Minasny *et al.* (2017) found that high relative gains can occur in soils with low initial SOC and during the first decades after improved management, while recognising that rates decline as new equilibria are approached. Long-term Rothamsted experiments provide a more stringent test: Poulton *et al.* (2018) showed that substantial SOC increases can be achieved with sustained organic inputs, but also that manure availability, land competition, nutrient requirements and economics constrain scalability. The defensible inference is that soil restoration can make a meaningful contribution to mitigation, not that every hectare can sustain a fixed proportional annual increase.

Scale also exposes carbon-source accounting problems. Importing manure, compost or crop residues to a project field can increase its SOC while simply relocating carbon that would otherwise have remained elsewhere. Genuine atmospheric removal requires additional photosynthetic capture or diversion from a counterfactual fate that would have produced greater net emissions. Similarly, increased residue retention may conflict with livestock feed, fuel or other uses. A field-level SOC balance is therefore necessary but insufficient for evaluating climate effectiveness.

These distinctions are summarised in Table 1. The table emphasises that carbon farming has at least four analytically separate components: carbon input, transformation and stabilisation, avoided loss, and credible accounting. Treating them as interchangeable leads to weak policy design.

Table 1. Biophysical concepts that determine the credibility of agricultural soil-carbon gains

Concept	What changes	Why it matters for carbon farming	Principal limitation or caution	Supporting references
Carbon input	Plant residues, roots, rhizodeposition or imported organic materials increase carbon entering soil.	Necessary first step for rebuilding depleted SOC; strongly linked to productivity and vegetation duration.	Input does not equal storage because much is respired; imported materials may transfer rather than create sequestration.	Lal, 2004; Paustian <i>et al.</i> , 2016
POM formation	Relatively coarse, partially decomposed plant-derived material accumulates.	Responsive indicator of recent management and substrate supply.	More vulnerable to decomposition and disturbance than strongly mineral-associated carbon.	Lavallee <i>et al.</i> , 2020
MAOM formation	Microbial products and dissolved compounds associate with mineral surfaces.	Often contributes to longer-lived SOC and links carbon storage to nutrient stoichiometry and mineralogy.	Finite mineral capacity and site-specific mineral reactivity constrain further accumulation.	Lavallee <i>et al.</i> , 2020; Six <i>et al.</i> , 2002
Physical protection	Aggregation and spatial inaccessibility limit decomposer access.	Can increase persistence without requiring chemically recalcitrant compounds.	Protection can be disrupted by disturbance and changes in wetting, drying or aggregation.	Lehmann & Kleber, 2015; Lehmann <i>et al.</i> , 2020
Carbon saturation / new equilibrium	Marginal SOC gain declines as storage capacity and inputs approach a new steady state.	Explains why early sequestration rates cannot be extrapolated indefinitely.	Saturation thresholds vary with soil and fraction; not a single universal ceiling.	Six <i>et al.</i> , 2002; Poulton <i>et al.</i> , 2018
Stock accounting	SOC concentration is converted to mass using depth, bulk density and coarse-fragment information.	Required to distinguish true sequestration from concentration or depth-distribution artefacts.	Bulk density changes and inconsistent sampling depths can bias estimated gains.	Poeplau <i>et al.</i> , 2017; Wendt & Hauser, 2013

Note. SOC = soil organic carbon; POM = particulate organic matter; MAOM = mineral-associated organic matter

4. Management Pathways for Building Soil Carbon

4.1 Reduced Tillage and No-Tillage: Soil Quality Benefit Does Not Guarantee Net Sequestration

Reduced tillage is one of the most widely promoted carbon-farming practices, partly because early syntheses reported higher SOC under no-tillage than conventional cultivation. West and Post (2002), using a global dataset of long-term experiments, found

average sequestration following conversion to no-tillage and more complex rotations. That evidence was influential, but subsequent analyses showed that depth distribution is critical. No-tillage leaves residues near the surface and reduces mixing, so carbon concentration commonly rises in shallow layers even when the profile stock changes little.

Powlson *et al.* (2014) argued that the mitigation potential of no-tillage had been overstated because many comparisons measured only the topsoil, while gains at the surface could be offset by smaller stocks below. A systematic review by Haddaway *et al.* (2017) similarly found that tillage effects are sensitive to sampling depth, duration and study design. Ogle *et al.* (2019) later demonstrated that climate and soil characteristics determine where no-till can store carbon and contribute to greenhouse-gas mitigation. Taken together, these studies do not support a universal 'no-till equals sequestration' rule.

This criticism should not be misread as an argument against conservation tillage. Reduced soil disturbance can lower erosion risk, improve surface structure, conserve moisture and reduce fuel use, and these benefits may be highly valuable for adaptation and soil quality. The critical distinction is between agronomic value and the magnitude of additional SOC stored through the whole profile. Carbon-farming protocols should reward measured net outcomes or conservatively modelled changes, not practice adoption alone.

4.2 Cover Crops and Diversified Rotations

Cover crops and diversified rotations have a stronger process basis for SOC accumulation because they increase the duration of living roots, annual biomass inputs and rhizosphere activity. Poeplau and Don (2015) estimated an average SOC increase of about 0.32 Mg C ha⁻¹ yr⁻¹ during the early decades after cover-crop adoption, while emphasising eventual saturation and variability. This is modest at the field scale but potentially significant when applied across suitable cropland. The principal advantage over imported organic amendments is that much of the carbon derives from additional in situ photosynthesis.

Diversity can also alter microbial and nutrient dynamics. McDaniel *et al.* (2014) found that adding crops to rotations increased total soil carbon and nitrogen, with larger responses when rotations included cover crops, and substantially increased microbial biomass. These effects support a mechanism in which greater temporal and functional diversity increases carbon input quality and quantity while improving nutrient capture. Yet performance is climate- and management-dependent: cover crops can reduce stored soil water before the cash crop in water-limited environments, and their carbon benefit may be small when establishment is poor or terminated early.

The most defensible role for cover crops is therefore as a multifunctional intervention rather than a carbon-only technology. Nitrogen-scavenging species can reduce nitrate losses; legumes can add biologically fixed nitrogen but may increase nitrous oxide risk under some conditions; mixtures can extend rooting niches but add seed and management costs. Carbon gains should be interpreted together with yield, water use and nitrogen balance.

4.3 Organic Amendments and Integrated Nutrient Inputs

Manure, compost and crop-residue additions can increase SOC because they deliver organic carbon directly while also supplying nutrients that stimulate plant production. Gross and Glaser (2021), synthesising field studies, reported a substantial mean increase in SOC stocks under manure application, with responses conditioned by initial SOC, soil texture, climate and sampling depth. Long-term experiments similarly show that sustained organic inputs can build SOC for decades, although the amounts of manure required to maintain high rates are not universally available (Poulton *et al.*, 2018).

The major accounting caveat is displacement. Manure is not newly fixed atmospheric carbon at the point of field application; its climate value depends on the counterfactual storage, treatment and land destination. Moving manure from one field to another can create a project-level stock increase without a corresponding regional removal. Composting can reduce labile carbon before application, while also stabilising material and affecting methane or nitrous oxide emissions during processing. Robust assessment therefore requires system boundaries that include amendment production and alternative uses.

Integrated organic and mineral nutrient management may nevertheless be agronomically superior to treating fertiliser and carbon inputs as substitutes. Wei *et al.* (2016), synthesising long-term Chinese experiments, found that combined organic amendments and fertilisers improved both crop yields and soil organic matter relative to less integrated strategies. Liu *et al.* (2023) likewise found positive SOC responses to balanced nutrient fertilisation, with the strongest effects varying by nutrient level and environment. The mechanism is partly indirect: adequate nitrogen, phosphorus and potassium support biomass production and microbial transformation, increasing the carbon available for stabilisation.

4.4 Biochar and Pyrogenic Carbon

Biochar differs from most carbon-farming inputs because pyrolysis transforms biomass into carbon forms with greater persistence than the original feedstock. Smith (2016) identified biochar as a potentially important negative-emissions option but emphasised land, biomass, energy and cost constraints. Joseph *et al.* (2021) showed that biochar effects depend strongly on feedstock, pyrolysis conditions, soil pH, nutrient status and ageing. Benefits can arise not only from storing pyrogenic carbon but also from changes in nutrient retention, pH, porosity and microbial processes.

A systematic review of global meta-analyses by Schmidt *et al.* (2021) concluded that agricultural biochar can improve several soil properties and crop outcomes, but response heterogeneity is substantial. The strongest agronomic responses often occur in acidic,

nutrient-poor or coarse-textured soils rather than uniformly across already fertile soils. Biochar should therefore be viewed as an engineered amendment whose performance is formulation- and site-specific, not as a generic black-carbon input.

The climate case also requires life-cycle discipline. Feedstock must be sourced without causing land-use change or depriving other beneficial uses; pyrolysis emissions and co-products should be accounted for; and stable carbon fractions should be quantified using defensible methods. When waste biomass would otherwise decompose rapidly or emit methane, biochar systems can offer a stronger net-removal case. When feedstock is harvested specifically or transported long distances, the balance becomes less favourable.

4.5 Agroforestry and Perennial Carbon Inputs

Agroforestry increases the temporal continuity and vertical distribution of carbon inputs through woody biomass, perennial roots and litter. Shi *et al.* (2018) found higher soil carbon stocks in agroforestry systems than in reference agricultural systems across a global meta-analysis, while highlighting differences among climates, soil depths and system types. Trees can also reduce erosion, alter microclimate and recycle nutrients from deeper layers, providing pathways through which carbon storage and resilience may reinforce one another.

Table 2. Critical comparison of major carbon-farming practices

Practice	Primary carbon pathway	Evidence for SOC gain	Key agronomic co-benefits	Main limitations / trade-offs	Supporting references
Reduced/no tillage	Lower disturbance; altered residue placement and aggregation.	Inconsistent for whole-profile stocks; surface gains are common.	Erosion control, fuel saving, surface structure and moisture conservation.	Depth redistribution can mimic sequestration; occasional tillage can reverse gains.	Haddaway <i>et al.</i> , 2017; Ogle <i>et al.</i> , 2019; Powlson <i>et al.</i> , 2014
Cover crops	Additional photosynthesis, roots and rhizodeposition between cash crops.	Generally positive but variable; mean rates are modest and decline toward saturation.	Nutrient capture, erosion control, biological activity, ground cover.	Water use, establishment cost, termination and N ₂ O risk depend on species and climate.	McDaniel <i>et al.</i> , 2014; Poeplau & Don, 2015
Diversified rotations	Greater residue diversity and temporal root activity.	Positive average effects on SOC and microbial biomass.	Break crops, nutrient cycling, pest management and yield stability.	Benefits depend on crop profitability, residue inputs and rotation length.	McDaniel <i>et al.</i> , 2014; West & Post, 2002
Manure/compost	Direct organic-C addition plus nutrient supply.	Often strong field-scale SOC increases.	N, P and micronutrient supply; aggregation; biological activity.	Carbon may be transferred rather than newly sequestered; nutrient surpluses and logistics.	Gross & Glaser, 2021; Poulton <i>et al.</i> , 2018
Integrated organic + mineral fertility	Higher biomass input and improved carbon-use efficiency under balanced nutrition.	Generally stronger than mineral fertiliser alone in long-term syntheses.	Yield, nutrient sufficiency and residue production.	Excess N can increase N ₂ O and environmental losses; optimum rates are site-specific.	Liu <i>et al.</i> , 2023; Wei <i>et al.</i> , 2016
Biochar	Addition of persistent pyrogenic C and alteration of soil chemistry/physics.	Potentially durable, with heterogeneous crop and SOC responses.	pH correction, nutrient retention and water/porosity benefits in responsive soils.	Feedstock sustainability, pyrolysis emissions, cost and variable product quality.	Joseph <i>et al.</i> , 2021; Schmidt <i>et al.</i> , 2021; Smith, 2016
Agroforestry	Perennial woody and root inputs; reduced erosion and deeper nutrient capture.	Meta-analytic evidence supports increased SOC in many systems.	Microclimate buffering, biodiversity, diversified products and erosion control.	Competition, delayed returns, permanence and land-tenure constraints.	Shi <i>et al.</i> , 2018

Note. SOC = soil organic carbon; N = nitrogen; N₂O = nitrous oxide; P = phosphorus

These benefits are not cost-free. Tree establishment can compete with crops for water or light, delay financial returns and complicate mechanisation. Reported SOC gains may also coincide with above-ground biomass carbon that should be separately accounted. Where land tenure is insecure or annual crop production is prioritised, adoption may be constrained even when long-term

ecosystem benefits are positive. The evidence therefore supports agroforestry most strongly in landscapes where perennial vegetation fits production goals and hydrological conditions.

Table 2 compares the main management pathways. The pattern across interventions is more informative than any single mean rate: practices that create additional plant carbon and correct local constraints generally have more defensible sequestration mechanisms than practices defined solely by disturbance reduction, while imported amendments require especially careful system-boundary accounting.

5. Carbon–Nutrient Coupling: Why Sequestration Cannot Be Managed as Carbon Alone

5.1 Nutrient Stoichiometry and Microbial Transformation

Persistent SOC formation is constrained by more than carbon supply. Microbial growth and the formation of microbial residues require nitrogen, phosphorus, sulphur and other nutrients, while plants require those same resources to produce the biomass that ultimately enters soil. In depleted systems, adding a carbon-rich residue without correcting nutrient limitation can increase decomposition but yield little stable carbon. Conversely, balanced fertilisation can raise crop production and residue return sufficiently to increase SOC even when the fertiliser itself contains no carbon (Liu *et al.*, 2023).

This coupling complicates narratives that portray mineral fertiliser as inherently antagonistic to carbon farming. Excess nitrogen can increase nitrous oxide emissions, acidification and nitrate loss, but severe nutrient deficiency can suppress biomass inputs and thereby limit SOC recovery. The relevant comparison is not organic versus mineral in isolation; it is whether nutrient supply is matched to crop demand, carbon inputs and soil buffering capacity. Long-term evidence from integrated nutrient management supports this interpretation because combinations of organic materials and mineral nutrients often outperform either component alone for sustained production and soil organic matter (Wei *et al.*, 2016).

The POM–MAOM framework adds a mechanistic explanation. Carbon entering POM can accumulate relatively quickly when residue inputs increase, whereas formation of MAOM often involves microbial processing and mineral surfaces. Nutrient availability can influence microbial carbon-use efficiency and the quantity of microbial products available for association with minerals. Thus, management that maximises residue biomass but induces strong nutrient imbalance may not maximise persistent carbon.

5.2 Nitrogen: Yield Support, Nitrous Oxide Risk and Carbon-Accounting Boundaries

Nitrogen is the most consequential nutrient for carbon-farming climate accounts because it can support SOC accumulation while simultaneously generating nitrous oxide, a potent greenhouse gas. A fertiliser-driven increase in crop biomass may raise soil carbon stocks, but the net climate benefit can be reduced or eliminated if nitrogen inputs substantially increase direct or indirect nitrous oxide emissions. Carbon-credit protocols that quantify SOC without including relevant changes in fertiliser-derived greenhouse gases therefore risk overstating mitigation.

Cover crops illustrate the trade-off. Non-legume species can retain residual mineral nitrogen that might otherwise leach, whereas legumes can reduce reliance on synthetic nitrogen through biological fixation. Yet rapid decomposition of N-rich residues can create periods of high mineral nitrogen availability. The climate outcome depends on termination timing, soil moisture, oxygen status and subsequent crop uptake. Management should therefore target synchrony between mineralisation and crop demand rather than simply maximise biomass.

Manure systems have an analogous problem. Manure can increase SOC strongly, as shown by Gross and Glaser (2021), but its nitrogen and phosphorus loads can exceed crop removal when applied primarily to meet carbon goals. That creates risks of nitrate leaching, phosphorus accumulation and gaseous nitrogen loss. Where manure is already generated, redistributing it according to nutrient need may improve regional efficiency, but carbon credit should not be granted merely because the receiving field gains SOC.

5.3 Phosphorus, Potassium and Soil Chemical Constraints

Phosphorus and potassium receive less attention in carbon accounting, yet they influence the productivity and microbial processes on which SOC accumulation depends. Liu *et al.* (2023) found that the SOC response to fertilisation varied with N, P and K combinations and environmental conditions, demonstrating that balanced nutrient supply can matter more than a single-nutrient prescription. In highly weathered tropical soils, phosphorus sorption and acidity can constrain plant growth and microbial processing; in other systems, potassium limitation can restrict biomass even where nitrogen is adequate.

Biochar can interact strongly with these constraints. Depending on feedstock and pyrolysis temperature, biochar may supply ash nutrients, raise pH or increase cation retention. Joseph *et al.* (2021) found especially strong plant responses where biochar formulations addressed specific nutrient or acidity constraints. This context dependence is important: a positive yield response to nutrient-rich biochar does not prove that stable carbon alone caused the agronomic benefit, and the same material may produce little response in a neutral, fertile soil.

The implication is that carbon farming needs soil testing and nutrient budgets, not a parallel carbon plan detached from fertility management. SOC targets should be embedded in crop nutrient strategies that measure nutrient inputs, removals and losses. This reduces the risk that carbon accumulation is achieved through inefficient nutrient use or environmental burden shifting.

Table 3 summarises the principal carbon–nutrient interactions. The practical message is that nutrient sufficiency can enable carbon storage, whereas nutrient excess can erode the net climate benefit; the optimum therefore depends on both SOC response and whole-system nutrient losses.

Table 3. Carbon–nutrient interactions relevant to carbon farming

Interaction	Potential benefit for SOC / productivity	Climate or environmental risk	Management implication	Supporting references
Balanced N supply	Supports crop biomass, roots and residue-C inputs; can increase SOC where N limits production.	Excess N increases N ₂ O risk, nitrate loss and acidification.	Optimise N to crop demand and include N ₂ O in climate accounting.	Liu <i>et al.</i> , 2023; Paustian <i>et al.</i> , 2016
P and K sufficiency	Removes non-N yield constraints and can improve carbon inputs and microbial processing.	Over-application can create nutrient imbalance and off-site loss, especially for P.	Use soil tests and balanced nutrient plans rather than carbon-driven fertiliser escalation.	Liu <i>et al.</i> , 2023
Manure/organic amendments	Direct C input plus multiple nutrients; frequently increases SOC.	N/P surplus, gaseous losses, transport burdens and transferred-carbon accounting.	Credit only net additional climate benefit; apply according to nutrient need and counterfactual fate.	Gross & Glaser, 2021; Poulton <i>et al.</i> , 2018
Crop residues and cover crops	In situ photosynthetic C and nutrient capture; supports microbial biomass.	High C:N residues may immobilise N; legume residues may create N ₂ O pulses.	Select species and termination to synchronise nutrient release with crop demand.	McDaniel <i>et al.</i> , 2014; Poeplau & Don, 2015
Biochar–nutrient interaction	Can retain nutrients, modify pH and improve nutrient-use efficiency in responsive soils.	Nutrient content and contaminant risk vary by feedstock; benefit can be misattributed to stable C.	Characterise product chemistry and target known soil constraints.	Joseph <i>et al.</i> , 2021; Schmidt <i>et al.</i> , 2021

Note. SOC = soil organic carbon; N = nitrogen; P = phosphorus; K = potassium; N₂O = nitrous oxide

6. Climate Resilience, Soil Function and Crop Performance

6.1 Soil Structure, Infiltration and Water Relations

The adaptation value of carbon farming rests partly on the relationship between SOC and soil physical function. Organic matter contributes to aggregate formation, pore continuity and resistance to crusting, which can improve infiltration and reduce erosion. Root-based carbon inputs are especially relevant because roots create biopores and stabilise aggregates within the profile. These processes can improve the ability of soil to capture intense rainfall and reduce runoff, benefits likely to become more important as precipitation becomes more variable.

Water-holding claims require more calibration. Minasny and McBratney (2018) showed that the direct effect of increasing organic matter on available water capacity is often limited, particularly compared with the influence of soil texture. This does not mean SOC is irrelevant to drought resilience. Improved aggregation, infiltration, rooting depth and surface cover can alter the effective water balance even if the laboratory water-retention curve changes little. The distinction between water storage capacity and field water availability explains why agronomic responses can be larger than would be predicted from organic matter concentration alone.

Reduced tillage and cover crops can therefore improve drought performance through multiple pathways that are not fully captured by SOC stock. Surface residues reduce evaporation, continuous pores aid infiltration and root channels facilitate deeper water access. Conversely, cover crops can deplete pre-plant soil moisture in dry regions. The net resilience effect is governed by rainfall timing, termination, crop rooting and soil depth rather than by the nominal practice itself.

6.2 Yield Level, Yield Stability and Soil Quality

SOC is frequently associated with higher yields, but causality is difficult to isolate because fertile, well-managed soils tend to have both higher SOC and higher productivity. Oldfield *et al.* (2019) found a positive global relationship between soil organic matter and crop yields, while also showing that the response is non-linear and context dependent. The agronomic benefit is greatest where organic matter alleviates a genuine constraint, such as nutrient retention, poor structure or water limitation, rather than in soils already functioning near optimum.

Evidence linking soil quality to resilience is more compelling when yield variability is analysed across climate conditions. Qiao *et al.* (2022), using more than 12,000 site-years from major cereal systems in China, found that high-quality soils were associated with higher average yields and lower sensitivity to climate variability. The study did not isolate SOC as a single causal factor, but it supports a systems interpretation: improving soil quality can buffer climatic stress. Carbon farming should therefore be evaluated using yield stability and downside-risk metrics as well as mean yield and SOC stock.

There is also an important temporal dimension. Practices that increase SOC may take years to generate measurable stock changes, while some resilience benefits appear sooner through surface cover, rooting or aggregation. Conversely, a measured SOC gain does not guarantee resilience if the cropping system remains dependent on shallow rooting, poorly timed nutrients or unsustainable irrigation. Carbon farming is most likely to improve resilience when it is part of a broader redesign that increases biological activity, rooting continuity and resource-use efficiency.

6.3 Resilience Is Context-Specific, Not a Universal Co-Benefit

The literature often bundles sequestration, fertility and resilience into a single package of co-benefits, but their spatial patterns need not coincide. A clay-rich temperate soil may gain SOC from manure but show little additional available water; a coarse, acidic tropical soil may respond strongly to biochar through pH and nutrient retention; a dryland soil may benefit from residue cover but lose water to an unmanaged cover crop. These examples show why benefit stacking must be demonstrated, not assumed.

Agroforestry can provide microclimate buffering and erosion control in addition to carbon storage, but competition for water may become important in dry environments. Likewise, reduced tillage can protect surface structure yet provide small whole-profile sequestration. The correct unit of analysis is therefore the management system in its biophysical context. Carbon-farming recommendations should specify the target constraint, expected mechanism and potential failure mode.

Table 4 distinguishes plausible resilience pathways from common overstatements. The evidence is strongest for improvements in soil structure, erosion protection and yield stability under some conditions; it is weaker for simple universal rules linking a fixed SOC increment to a fixed amount of plant-available water or climate-proofing.

Table 4. Soil-carbon pathways to climate resilience and their evidentiary limits

Resilience pathway	Evidence-supported mechanism	What can reasonably be expected	Why outcomes vary	Supporting references
Aggregation and infiltration	Organic binding agents, roots and microbial products promote aggregate stability and pore continuity.	Reduced crusting/erosion and improved rainfall capture in responsive soils.	Texture, tillage history, sodicity, rainfall intensity and residue cover modify the response.	Lal, 2004; Paustian <i>et al.</i> , 2016
Available water	Organic matter alters pore-size distribution and water retention.	Modest direct gains in many mineral soils; indirect field benefits may be larger via structure and roots.	Mineral texture and baseline organic matter dominate many water-retention responses.	Minasny & McBratney, 2018
Nutrient buffering	SOC and amendments increase exchange capacity, microbial nutrient cycling and nutrient retention.	Greater nutrient-use efficiency where nutrient loss or low retention limits yield.	pH, mineralogy and nutrient form determine whether added C improves or immobilises nutrients.	Joseph <i>et al.</i> , 2021; Liu <i>et al.</i> , 2023
Yield stability	Higher soil quality can reduce sensitivity of crop yields to climatic variability.	Lower downside risk and more stable yields in some cropping systems.	SOC is one component of soil quality; irrigation, cultivar, rooting and management co-determine outcomes.	Oldfield <i>et al.</i> , 2019; Qiao <i>et al.</i> , 2022
Microclimate / perennial buffering	Trees and perennial cover reduce wind, erosion and temperature extremes.	Potential improvement in resilience and diversified production.	Competition for water/light, system design and land tenure can offset benefits.	Shi <i>et al.</i> , 2018

Note. SOC = soil organic carbon

7. Net Climate Effectiveness: Trade-offs, Saturation and Reversibility

7.1 Sequestration Is Finite and Reversible

A central weakness in some carbon-farming narratives is the treatment of SOC accumulation as a permanent linear flow. In reality, sequestration rates decline as a new equilibrium is approached, and previously stored carbon can be lost when management changes, drought or erosion accelerates decomposition, or land use shifts. Smith (2016) identified saturation and reversibility as fundamental constraints on soil-carbon negative emissions. These are not minor accounting details; they distinguish temporary biological storage from geological carbon storage.

Long-term experiments make the point empirically. Poulton *et al.* (2018) observed substantial increases under repeated manure inputs, but rates changed over time and depended on continued management. Minasny *et al.* (2017) similarly showed that high relative annual increases are most plausible in low-carbon soils during earlier phases of restoration. A carbon contract that assumes the initial rate persists over the crediting period can therefore issue more credits than the soil can sustain.

Permanence is also behavioural. A farmer may adopt a carbon-building practice for the duration of a payment and later reverse it because of crop economics, weed control or land transfer. Effective programmes need buffers, reversal liability or long-term stewardship arrangements. The soil science alone cannot guarantee persistence without institutions that align incentives over time.

7.2 Whole-System Greenhouse-Gas Accounting

SOC stock change is only one component of climate impact. A practice can increase SOC while also increasing nitrous oxide, methane or fossil-energy emissions. Organic amendments may generate emissions during storage and transport; fertiliser used to support biomass input can raise nitrous oxide; irrigation required to produce additional residues consumes energy and water. Conversely, reduced tillage may lower fuel use even when its SOC benefit is small. A credible assessment therefore reports carbon dioxide removals and changes in non-CO₂ greenhouse gases separately before converting them to a net climate balance.

Leakage extends the accounting boundary further. Diverting crop residues to one field may reduce feed availability or soil inputs elsewhere. Allocating land to biomass production for biochar can displace food production or cause land-use change. These effects are difficult to measure but cannot be ignored when scaling a project-based result to regional climate policy. The strongest carbon-farming strategies use additional in situ biomass, wastes with low-value counterfactuals, or practices that improve resource-use efficiency without creating displacement.

The implication is not that every project requires a perfect global life-cycle model. Rather, protocols should identify the largest plausible off-site effects and apply conservative assumptions where direct measurement is impossible. Carbon accounting should become stricter as the claim becomes more fungible: a practice promoted for soil health can tolerate uncertainty that would be unacceptable if its credits are used to offset fossil emissions.

7.3 The Appropriate Role of Carbon Farming in Mitigation Portfolios

Soil carbon is valuable partly because it can deliver benefits while agriculture continues to produce food. That makes it attractive in near-term mitigation portfolios, especially in degraded croplands. Yet the finite and reversible nature of the sink means it cannot justify delayed decarbonisation. Powlson *et al.* (2011) cautioned against conflating technically possible SOC increases with large permanent offsets, while Bossio *et al.* (2020) placed soil carbon within a wider portfolio of natural climate solutions that includes protection as well as restoration.

A hierarchy emerges from the evidence. Avoiding losses from high-carbon soils and erosion can prevent emissions that may be difficult to reverse. Rebuilding severely depleted mineral soils can deliver high agronomic value and meaningful but finite removals. Maintaining already restored SOC is a stewardship function even when it produces no new annual sequestration. Carbon markets designed only around annual increments may undervalue this maintenance stage while over-rewarding early, uncertain gains.

Carbon farming is therefore best framed as a transition pathway: restore depleted soils, improve management efficiency and then maintain the improved state, while energy and industrial emissions are reduced independently. This framing aligns climate integrity with agronomic reality and reduces pressure to promise indefinite sequestration.

8. Measurement, Reporting, Verification and Carbon-Market Integrity

8.1 Measuring Small Changes Against Large and Variable Stocks

Measurement, reporting and verification (MRV) is unusually difficult for SOC because the annual change being claimed is often small relative to the existing stock and spatial variability within a field. Smith (2004) showed that detectable change depends on sampling intensity, variability, expected rate and time interval. Short crediting periods create pressure to infer sequestration from models or practice adoption before a statistically robust stock change can be measured.

Bulk density is a frequent source of bias. Carbon concentration can rise because soil becomes less dense, while the amount of carbon per unit land area changes less. Poeplau *et al.* (2017) showed that misuse of bulk density and rock-fragment data can systematically overestimate SOC stocks. Equivalent soil-mass approaches can reduce errors when treatments change compaction or soil volume, and Wendt and Hauser (2013) provided a practical procedure for monitoring across multiple layers.

Sampling depth, geolocation and resampling design also matter. If a no-till project samples only the upper 10 or 15 cm, it may capture redistribution rather than whole-profile sequestration. Fixed-depth sampling can be retained for operational consistency, but equivalent mass calculations and deeper layers are needed where treatment-induced density or depth redistribution is plausible. Archived samples improve quality assurance and allow later re-analysis as methods evolve.

8.2 Additionality, Baselines, Permanence and Leakage

Carbon credits require more than accurate measurement. Additionality asks whether the SOC increase would have occurred without the incentive. A farmer already using cover crops should not necessarily generate new credits simply by registering an existing practice, yet excluding early adopters can create perverse incentives. Baselines must therefore combine historical management, regional practice prevalence and credible counterfactual modelling rather than rely on self-declared intent.

Permanence is more difficult because SOC remains biologically active. Murray *et al.* (2007) demonstrated that permanence, leakage and additionality materially affect the economics of soil-carbon sequestration projects. Temporary storage has climate value, but it is not equivalent to permanent removal unless the accounting framework explicitly addresses duration and reversal. Buffers, discounting and pooled risk mechanisms can reduce over-crediting but cannot eliminate uncertainty.

Oldfield *et al.* (2022) argued that regional consistency is essential for agricultural soil-carbon crediting. This is important because protocols that use different baselines, depths, models or uncertainty deductions can issue different numbers of credits for similar biophysical outcomes. Fragmented standards also create incentives to select the most permissive methodology. Harmonisation should focus on core principles—stock-change definition, depth, uncertainty, additionality and reversal treatment—while retaining flexibility for region-specific sampling and models.

8.3 From Practice-Based Incentives to Outcome-Based Stewardship

Practice-based payments are administratively simple but can mistake average effects for local outcomes. The evidence reviewed here shows why that is risky: no-till is not universally sequestration-positive; cover-crop performance depends on biomass and water; manure may transfer carbon; biochar quality varies; and SOC responses decline with time. Pure outcome-based crediting is scientifically stronger but expensive because repeated soil sampling must detect small changes over heterogeneous fields.

A hybrid approach is more defensible. Programmes can use practice information and validated models to stratify expected response, require direct measurement on a statistically designed subset, and update model parameters as field data accumulate. Conservative uncertainty deductions can prevent issuing credits for noise. The objective should be to minimise expected over-crediting rather than to manufacture false precision.

The strongest criticism of current offset-oriented models comes from Paul *et al.* (2023), who concluded that private soil-carbon certificates face unresolved problems of permanence, additionality, monitoring, leakage and accountability. That conclusion does not negate financial support for carbon-building practices. It instead suggests that payments for ecosystem services, transition grants or stewardship contracts may sometimes fit the biophysical product better than one-for-one offset credits. Soil restoration can be worth funding even when its carbon is not sufficiently certain or durable to neutralise a fossil emission.

8.4 Adoption, Transaction Costs and Distributional Consequences

A technically effective carbon-farming practice can fail at scale if the incentive does not cover transition cost, risk and management complexity. Farmers make decisions under uncertainty about weather, commodity prices, land tenure and future policy, whereas carbon contracts may require long commitments for a revenue stream that is small relative to crop income. This asymmetry is especially important for cover crops, agroforestry and diversified rotations, where establishment costs or short-term yield risks occur before a measurable carbon benefit. Carbon programmes that treat adoption as a simple response to a uniform price therefore overlook the opportunity cost of changing an operating farm system.

Transaction cost is closely connected to MRV stringency. Direct soil measurement improves confidence but requires sampling, laboratory analysis, data management and verification. When expected sequestration per hectare is modest, these fixed costs can absorb a large fraction of the carbon payment. Model-based approaches can reduce expense, but only by shifting uncertainty from sampling to model assumptions. The appropriate response is not to lower evidentiary standards until projects become profitable; it is to design aggregation, stratified sampling and public measurement infrastructure that spreads the cost across many farms while preserving conservative crediting.

Distributional effects also influence permanence. Tenants may be unwilling to invest in practices whose soil benefits accrue after a lease ends, while owners may capture part of the value through land appreciation. Smallholders may have limited access to seed, machinery or biomass even where the biophysical response would be high. Agroforestry can be constrained by insecure tree tenure or delayed revenue. These barriers mean that the locations with the largest carbon-restoration potential are not automatically the locations where market-based adoption is easiest. Public programmes may need to combine carbon objectives with extension, credit access and risk sharing rather than rely on offset revenue alone.

The design of incentives should also avoid penalising prior stewardship. Strict project-level additionality can exclude farmers who adopted good practices before a carbon programme, creating an equity problem and a perverse signal to late adopters. At the same time, paying full removal credits for long-established practices would overstate new atmospheric benefit. A defensible distinction is to separate payments for new verified sequestration from stewardship payments for maintaining valuable SOC and ecosystem services. Such separation recognises that preserving an improved soil state is socially valuable even when it no longer generates additional annual removal.

Carbon farming is therefore partly an institutional coordination problem. Scientific uncertainty, farmer risk and verification cost interact: a programme can maximise none of them independently. High-integrity implementation requires a payment structure that is large enough to change behaviour, a monitoring system capable of detecting meaningful outcomes, and contractual arrangements that survive changes in operators or market conditions. Where those conditions cannot be met, soil-health support may remain justified, but the intervention should not be sold as a fungible tonne-for-tonne offset.

8.5 Organic Farming, Regenerative Labels and the Limits of Practice Bundles

Carbon farming is often discussed alongside organic, conservation and regenerative agriculture, but these labels are not equivalent to measured carbon outcomes. Organic systems can increase SOC because they commonly rely on manure, compost, legumes and diversified rotations. Gattinger *et al.* (2012) reported higher topsoil carbon stocks under organic farming in a meta-analysis, but interpretation requires attention to external organic inputs, system productivity and the land area needed to produce equivalent output. A production label can therefore identify a management philosophy without specifying the direction or magnitude of net greenhouse-gas change.

Regenerative agriculture presents a similar evidentiary challenge because the term covers varying combinations of reduced tillage, cover crops, livestock integration, diverse rotations and input reduction. Some of these components have strong evidence for soil-function benefits; others produce highly context-dependent carbon changes. Aggregating them under a positive label can obscure which mechanism generated the observed outcome. For research and policy, the more useful unit is the specific management transition and its measured consequences rather than membership in a broad farming category.

This distinction becomes important when carbon markets use practice eligibility as a proxy for sequestration. A farm adopting several recognised practices may still show little measurable SOC gain if biomass production is low, baseline carbon is already high or drought limits cover-crop growth. Conversely, a conventional farm that improves nutrient balance and substantially increases residue return may build SOC without carrying a regenerative label. Outcome claims should therefore remain tied to verified biophysical change, while broader sustainability certification may consider a wider set of practices and values.

8.6 Spatial Targeting and the Value of Avoided Soil-Carbon Loss

The carbon-farming debate often emphasises additional storage, yet preventing loss from existing stocks can be equally important. Agricultural expansion and degradation have created a large historical SOC debt, with substantial spatial variation in the magnitude of depletion (Sanderman *et al.*, 2017). This variation implies that restoration effort should be targeted. Depleted croplands with sufficient water, nutrient supply and mineral capacity may offer a stronger combination of sequestration and productivity benefits than soils already near a management-induced equilibrium (Amelung *et al.*, 2020).

Table 5. Measurement and governance requirements for credible carbon farming

Integrity issue	Failure mode	Preferred safeguard	Residual uncertainty	Supporting references
SOC stock measurement	Concentration changes or bulk-density shifts are misread as stock change.	Measure bulk density/coarse fragments; use equivalent soil mass where appropriate; repeat georeferenced sampling.	Fine-scale spatial heterogeneity and laboratory error remain.	Poeplau <i>et al.</i> , 2017; Wendt & Hauser, 2013
Sampling depth	Surface accumulation is credited despite offsetting deeper change.	Pre-specify depth; include deeper layers when management redistributes C or roots.	Subsoil sampling increases cost and variance.	Powlson <i>et al.</i> , 2014; Wendt & Hauser, 2013
Detectability	Short monitoring interval cannot distinguish treatment effect from spatial noise.	Power-based sampling design and multi-year intervals; combine measurements with calibrated models.	Slow SOC change can remain uncertain for years.	Smith, 2004
Additionality	Credits pay for business-as-usual practices or inevitable changes.	Historical management records, regional adoption benchmarks and conservative baselines.	Counterfactual behaviour is inherently uncertain.	Oldfield <i>et al.</i> , 2022; Paul <i>et al.</i> , 2023
Permanence	SOC is later re-emitted after management reversal or disturbance.	Long-term contracts, buffers, reversal liability and discounted crediting.	Biological storage cannot be made fully irreversible.	Murray <i>et al.</i> , 2007; Paul <i>et al.</i> , 2023
Leakage / transfer	Residues, manure or land-use pressures shift emissions outside the project boundary.	Track major biomass sources and displaced uses; apply system-level accounting.	Indirect market-mediated effects can be difficult to quantify.	Murray <i>et al.</i> , 2007; Paul <i>et al.</i> , 2023
Protocol consistency	Similar projects receive different credits because standards use different depths or assumptions.	Harmonise minimum MRV principles and uncertainty deductions regionally.	Local models still need site-specific calibration.	Oldfield <i>et al.</i> , 2022

Note. SOC = soil organic carbon; MRV = measurement, reporting and verification; C = carbon

Targeting should also recognise areas where protection dominates restoration. High-carbon organic soils and erosion-prone fields can release large amounts of stored carbon when disturbed, although organic-soil conservation lies partly outside the central mineral-soil scope of this review. Within croplands, erosion control can avoid the displacement and mineralisation of carbon while

preserving productive topsoil. Programmes focused exclusively on newly measured stock increases may under-reward these avoided-loss functions because prevention does not appear as a positive annual sequestration rate.

Spatial opportunity maps can guide investment, but they should not be converted directly into credit baselines. Global datasets smooth local variation and may lack management histories, bulk-density measurements or subsoil observations. Their strongest use is prioritisation: identify regions where historic depletion, yield gaps and feasible management changes overlap, then establish field-scale baselines using local measurements. This nested approach retains the strategic value of global analyses without pretending that coarse maps provide project-level verification.

The policy consequence is a shift from universal practice incentives towards diagnosis-based carbon farming. A low-carbon soil caused by erosion needs ground cover and structural protection; a low-carbon soil caused by low productivity may need balanced fertility and water management; a coarse acidic soil may respond to suitable organic inputs or biochar; a soil near saturation may merit maintenance rather than aggressive sequestration targets. Such targeting uses SOC as a state variable within agronomy rather than as an isolated commodity.

Table 5 translates the scientific limitations into MRV requirements. The central principle is conservative comparability: credits should reflect a lower-confidence bound on net additional climate benefit, whereas soil-health programmes can reward verified management and co-benefits without claiming equivalent permanent offsets.

9. Future Directions

The immediate research priority is to replace broad practice labels with response functions that predict where, when and for how long SOC will increase. Multi-site experiments should stratify soils by texture, mineralogy, baseline SOC, climate and production system and should measure whole-profile stocks using common depth and equivalent-mass protocols. Repeated sampling over at least decadal horizons is necessary to estimate saturation trajectories and reversal risks rather than only initial response. Fractionation into POM and MAOM should be incorporated in representative sites to determine whether management is increasing responsive carbon, persistent carbon or both. This would materially improve model transferability and help distinguish soil-health gains from durable climate removal.

A second priority is integrated carbon–nutrient greenhouse-gas experimentation. Trials should report SOC stock change together with crop yield, nitrogen and phosphorus balances, nitrous oxide, relevant methane fluxes and major energy inputs. This is particularly important for manure, compost, legume cover crops and fertiliser-supported carbon strategies. The unresolved question is not whether these practices can build SOC, but under which nutrient regimes their net greenhouse-gas balance remains favourable. Factorial designs that vary carbon input and nutrient supply can identify thresholds at which extra biomass ceases to justify additional nitrogen loss.

Third, climate-resilience research should move beyond average yield. Long-term datasets and coordinated field networks should quantify yield variance, downside risk during drought or heat, infiltration, rooting depth, erosion and recovery after extreme events. Qiao *et al.* (2022) demonstrated the value of large site-year datasets for linking soil quality with climate sensitivity, but comparable analyses are needed across drylands, humid tropics and rainfed temperate systems. Causal mediation analysis can help separate the contribution of SOC from correlated improvements in fertility, structure and management intensity.

Fourth, biochar research requires tighter linkage between product characterisation and field response. Trials should report feedstock origin, pyrolysis conditions, stable-carbon metrics, ash nutrients, pH, contaminants and life-cycle emissions. Rather than asking whether 'biochar works', experiments should predict which formulation corrects which soil constraint and quantify the persistence of both pyrogenic carbon and any induced change in native SOC. This would shift the field from average treatment effects toward engineering specifications suitable for agronomic and carbon accounting.

Fifth, MRV research should be designed around decision thresholds. The relevant question for a crediting programme is whether the lower bound of a measured or modelled net benefit exceeds the quantity credited. Sampling optimisation, remote sensing, proximal soil sensing and process models should therefore be evaluated on error propagation and over-crediting risk, not only correlation with laboratory SOC. Open benchmark datasets containing repeated field measurements, management histories and uncertainty metadata would allow independent comparison of models and discourage proprietary black-box estimates.

Sixth, carbon-market governance should test alternative payment architectures. Paul *et al.* (2023) identify reasons why temporary, uncertain SOC increases fit poorly into permanent-offset logic. Comparative policy experiments should evaluate results-based stewardship payments, practice-transition grants, public procurement, insurance-linked incentives and carbon credits with explicit ton-year or reversal discounting. Outcomes should include farmer participation, transaction costs, distributional equity, long-term practice retention and verified net greenhouse-gas benefit. This is particularly important for smallholders, for whom high MRV costs can consume much of the potential payment.

A final priority is regional targeting. Global maps identify broad opportunity, but implementation needs locally validated baselines and constraint diagnoses. Research should prioritise degraded and carbon-depleted croplands where SOC restoration is likely to improve productivity or resilience, while avoiding blanket incentives on soils already near equilibrium. Decision-support systems should combine soil carbon status, yield gaps, water limitation, nutrient balance, erosion risk and biomass availability. Such targeting would improve both climate efficiency and agronomic relevance, aligning the strongest carbon-farming investments with sites where multiple constraints can be addressed together.

10. Conclusions

Carbon farming can contribute to sustainable agriculture when it is treated as management of a coupled soil–plant–nutrient system rather than as a simple practice checklist. The strongest evidence supports rebuilding depleted SOC by increasing in situ plant inputs, diversifying rotations, maintaining cover, using organic amendments judiciously, integrating nutrients to sustain biomass production, introducing agroforestry where agronomically compatible, and using well-characterised biochar in responsive soils. The evidence is weaker for universal mitigation claims based on no-tillage or any other single practice without whole-profile stock measurement. SOC accumulation is finite, site-specific and reversible, and apparent gains can be distorted by sampling depth, bulk density or transfer of organic material between locations. Nutrient management is integral because nitrogen, phosphorus and potassium can enable biomass production and microbial carbon stabilisation, while nutrient excess—especially nitrogen—can create greenhouse-gas and water-quality costs that offset part of the climate benefit. Climate resilience is a credible co-benefit where carbon-building practices improve structure, infiltration, rooting, nutrient buffering and yield stability, but the magnitude varies strongly among soils and production systems. Carbon farming should therefore complement, not substitute for, direct emissions reduction. Its most defensible policy role is targeted soil restoration and long-term stewardship, with carbon credits reserved for cases in which additionality, net greenhouse-gas benefit, uncertainty, permanence and leakage are handled conservatively. This framing preserves the substantial agronomic value of better soil management without asking biological carbon storage to provide a degree of permanence or measurement certainty that the evidence does not support.

11. Limitations

As a critical narrative review, this synthesis is intrinsically vulnerable to selection and interpretive bias despite a transparent search and appraisal process. The evidence base is heterogeneous in experimental duration, sampling depth, SOC measurement, baseline fertility, climate, crop system and outcome reporting, which limits direct comparison of sequestration rates among practices. Some influential global assessments depend on models and scenarios rather than long-term direct measurement, while long-term field experiments are geographically concentrated in temperate regions and may not represent tropical, dryland or smallholder systems adequately. English-language and accessibility constraints may have reduced representation of some regional literature. Quantitative meta-analysis was not undertaken, so effect sizes reported from existing syntheses were interpreted in their original methodological context rather than recombined. Rapid development of carbon-market standards also means that governance practices may evolve after the literature search end date. These limitations reinforce the need to interpret the review as an evidence-calibrated synthesis of mechanisms, consistency and implementation constraints rather than as an exhaustive inventory of all carbon-farming studies.

Declaration of AI Use

This manuscript was prepared through the intellectual contributions of the author(s) to the study design, data analysis, interpretation of results, and scholarly content. Grammarly and ChatGPT were used solely to assist with grammatical refinement and reference formatting during manuscript revision. The author(s) accept full responsibility for the accuracy, integrity, and final content of the manuscript.

Competing Interests

Authors have declared that no competing interests exist.

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